

MSc

Course title: SM6 Bayes Methods
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1. For $i = 1, \dots, n$ let $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,p})$, $\mathbf{x}_i \in \mathbb{R}^p$ be a vector of p continuous covariates for the i 'th observation. Let $\mathcal{M}$ be the set of all subsets of $\{1, 2, \dots, p\}$ (including the empty set) and let $m \in \mathcal{M}$ be one such set. Let $\beta = (\beta_1, \dots, \beta_p)$, $\beta \in \mathbb{R}^p$ be a vector of regression parameters. For $m \in \mathcal{M}$, let $K = |m|$ give the number of elements in m . The labels in m are ordered so if $m = \{m_1, \dots, m_K\}$ then $m_1 < m_2 < \dots < m_K$. Denote by $\beta_m = (\beta_{m_1}, \dots, \beta_{m_K})$ the vector of parameters in the model when $|m| = K$. For $m \in \mathcal{M}$ with $K = |m|$, let

$$y_i = \sum_{j=1}^K \beta_{m_j} x_{i,m_j} + \epsilon_i,$$

with $\epsilon_i \sim N(0, \sigma^2)$ jointly independent for $i = 1, \dots, n$, be the observation model for $y = (y_1, \dots, y_n)$ given the unknown true model is model $M = m$. Here $y_i = \epsilon_i$, $i = 1, \dots, n$ if $m = \emptyset$ so $K = 0$. Here $(\beta_m, \sigma) \in \Omega_m$ where $\Omega_m = \mathbb{R}^K \times (0, \infty)$ and $K = |m|$. A prior

$$\pi(\beta_m, \sigma | m) = \pi_\sigma(\sigma) \prod_{j=1}^K \pi(\beta_{m_j})$$

is given. The prior $\pi(\beta_m, \sigma | m)$ is a probability density on Ω_m . A model prior probability mass function $\pi_M(m)$, $m \in \mathcal{M}$ is given.

(a) [10 marks]

- (i) Write down the fixed-model posterior $\pi(\beta_m, \sigma | y, m)$ in terms of the normal observation model density and priors.

Solution: (2, standard)

$$\pi(\beta_m, \sigma | y, m) = \frac{p(y | \beta_m, \sigma, m) \pi(\beta_m, \sigma | m)}{p(y | m)}$$

where $\pi(\beta_m, \sigma | m)$ is given,

$$p(y | \beta_m, \sigma, m) = \prod_{i=1}^n N(y_i; \mathbf{x}_{i,m}^T \beta_m, \sigma^2)$$

in terms of let $\mathbf{x}_{i,m} = (x_{i,m_1}, \dots, x_{i,m_K})$, and

$$p(y | m) = \int_{\Omega_m} p(y | \beta_m, \sigma, m) \pi(\beta_m, \sigma | m) d\beta_m d\sigma.$$

- (ii) Give the joint posterior distribution for the model and parameter, $\pi(\beta_m, \sigma, m | y)$, in terms of the model elements, and the state space for (β_m, σ, m) .

Solution: (3, standard) We can write

$$\pi(\beta_m, \sigma, m | y) = \frac{p(y | \beta_m, \sigma, m) \pi(\beta_m, \sigma | m) \pi_M(m)}{p(y)}$$

with everything as above and

$$p(y) = \sum_{m \in \mathcal{M}} p(y | m) \pi_M(m).$$

The random variable $(\beta_m, \sigma, m) \in \Omega$ where

$$\Omega = \bigcup_{m \in \mathcal{M}} \bigcup_{(\beta_m, \sigma) \in \Omega_m} \{(\beta_m, \sigma, m)\}.$$

- (iii) Let model $m^* = \{1, \dots, p\}$ be the full model and suppose that, for $m \in \mathcal{M}$, estimates $\hat{B}_{m,m^*}$ of ratios $B_{m,m^*} = p(y|m)/p(y|m^*)$ are available. For $m \in \mathcal{M}$ and $t = 1, \dots, T$, let $\beta^{(m,t)}, \sigma^{(m,t)}$, $t = 1, \dots, T$ be samples distributed according to $\pi(\beta_m, \sigma|y, m)$. Let $g(\beta_m, \sigma, m)$ be a given function of the parameters and model index. Give an estimator for the model-averaged posterior expectation $E_{\beta_m, \sigma, m|y}(g(\beta_m, \sigma, m))$ in terms of $\hat{B}_{m,m^*}$, $m \in \mathcal{M}$ and the samples.

Solution: (5, new) We have

$$\pi(\beta_m, \sigma, m|y) = \pi(\beta_m, \sigma|y, m)\pi(m|y)$$

with $\pi(m|y) = p(y|m)\pi_M(m)/p(y)$ so we can write μ_g partitioning on m ,

$$\begin{aligned} E_{\beta_m, \sigma, m|y}(g(\beta_m, \sigma, m)) &= \sum_{m \in \mathcal{M}} E_{\beta_m, \sigma|y, m}(g(\beta_m, \sigma, m)) \frac{p(y|m)\pi_M(m)}{p(y)} \\ &= \sum_{m \in \mathcal{M}} E_{\beta_m, \sigma|y, m}(g(\beta_m, \sigma, m)) \frac{p(y|m)\pi_M(m)}{\sum_{\tilde{m}} p(y|\tilde{m})\pi_M(\tilde{m})} \\ &= \sum_{m \in \mathcal{M}} E_{\beta_m, \sigma|y, m}(g(\beta_m, \sigma, m)) \frac{B_{m,m^*}\pi_M(m)}{\sum_{\tilde{m}} B_{\tilde{m},m^*}\pi_M(\tilde{m})} \\ &= \sum_{m \in \mathcal{M}} E_{\beta_m, \sigma|y, m}(g(\beta_m, \sigma, m)) w_m \end{aligned}$$

where

$$w_m = \frac{B_{m,m^*}\pi_M(m)}{\sum_{\tilde{m}} B_{\tilde{m},m^*}\pi_M(\tilde{m})}.$$

Now let

$$\hat{w}_m = \frac{\hat{B}_{m,m^*}\pi_M(m)}{\sum_{\tilde{m}} \hat{B}_{\tilde{m},m^*}\pi_M(\tilde{m})}.$$

Our estimate for $\mu_g = E_{\beta_m, \sigma, m|y}(g(\beta_m, \sigma, m))$ is

$$\hat{\mu}_g = \sum_{m \in \mathcal{M}} \hat{w}_m \left[\frac{1}{T} \sum_{t=1}^T g(\beta_m^{(m,t)}, \sigma^{(m,t)}, m) \right].$$

(b) [10 marks]

- (i) For $c > 0$ let $(\rho_{m,m'})_{m,m' \in \mathcal{M}}$ be a matrix satisfying $\rho_{m,m'} = c$ if m and m' differ by exactly one element (and otherwise equal zero), so if

$$m \Delta m' = (m \setminus m') \cup (m' \setminus m)$$

then

$$\rho_{m,m'} = \begin{cases} c & \text{if } |m \Delta m'| = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Calculate c to ensure $\sum_{m'} \rho_{m,m'} = 1$ for every $m \in \mathcal{M}$.

Solution: (3, new) For every $m \in \mathcal{M}$ there are always just p models $m' \in \mathcal{M}$ satisfying $|m \Delta m'| = 1$ so $\rho_{m,m'} = 1/p$ for every $m, m' \in \mathcal{M}$ such that $|m \Delta m'| = 1$.

- (ii) Construct the dimension-changing updates of a reversible jump MCMC algorithm targeting $\pi(\beta_m, \sigma, m|y)$, including an update adding a component to β_m and an update deleting a component from β_m , with acceptance probabilities given in terms of the model elements. Use ρ to propose the new model.

Solution: (7, new) Here is the dimension-changing update of a RJ-MCMC sampler: Suppose $X_t = (\beta_m, \sigma, m)$.

Step 1. Choose $m' \in \mathcal{M}$ wp $\rho_{m,m'}$. Let $K = |m|$ and $K' = |m'|$.

Step 2(up). If $K' = K + 1$ with $i = m' \setminus m$, then we are proposing to add the effect β_i to the state.

Set $\beta'_{m'} = (\beta'_{m'_1}, \dots, \beta'_{m'_{K'}})$ with $\beta'_j = \beta_j$ for $j \in m$ and $\beta'_i = u$ with $u \sim g_{m,m'}(\cdot)$, where $g_{m,m'}(u) = \pi(u)$ is the single-component prior given in the question. In this case the acceptance probability is

$$\begin{aligned} \alpha(\beta'_{m'}, \sigma, m' | \beta_m, \sigma, m) &= \min \left\{ 1, \frac{\pi(\beta'_{m'}, \sigma, m' | y) \rho_{m',m}}{\pi(\beta_m, \sigma, m | y) \rho_{m,m'} g_{m,m'}(u)} \left| \frac{\partial(\beta'_{m'})}{\partial(\beta_m, u)} \right| \right\} \\ &= \min \left\{ 1, \frac{p(y | \beta'_{m'}, \sigma, m') \pi(\beta'_{m'}, \sigma | m') \pi_M(m')}{p(y | \beta_m, \sigma, m) \pi(\beta_m, \sigma | m) \pi_M(m) \pi(\beta'_i)} \right\} \\ &= \min \left\{ 1, \frac{p(y | \beta'_{m'}, \sigma, m') \pi_M(m')}{p(y | \beta_m, \sigma, m) \pi_M(m)} \right\} \end{aligned}$$

where the first line is taken from notes, the second line follows because the mapping from β_m, u to $\beta'_{m'}$ sets each component of $\beta'_{m'}$ equal exactly one of the elements of (β_m, u) , so the Jacobian factor must equal one, and the third line follows as $\pi(\beta'_{m'}, \sigma | m') = \pi(\beta_m, \sigma | m) \pi(\beta'_i)$.

Step 2(down). If $K' = K - 1$ with $i = m \setminus m'$, then we are proposing to delete the effect β_i from the state.

Set $\beta'_{m'} = (\beta'_{m'_1}, \dots, \beta'_{m'_{K'}})$ with $\beta'_j = \beta_j$ for $j \in m'$. In this case the acceptance probability has the same form

$$\alpha(\beta'_{m'}, \sigma, m' | \beta_m, \sigma, m) = \min \left\{ 1, \frac{p(y | \beta'_{m'}, \sigma, m') \pi_M(m')}{p(y | \beta_m, \sigma, m) \pi_M(m)} \right\},$$

using the fact that the Hastings ratio is the inverse with primes swapped. Note that now the model in the numerator is the smaller model etc.

Step 3 (accept/reject) Accept the new state and set $X_{t+1} = (\beta'_{m'}, \sigma, m')$ with probability $\alpha(\beta'_{m'}, \sigma, m' | \beta_m, \sigma, m)$ and otherwise set $X_{t+1} = (\beta_m, \sigma, m)$.

2. (a) [9 marks] Let $X_1, X_2, X_3, \dots$ be an infinite exchangeable sequence of binary random variables. For $n = 1, 2, 3, \dots$ and $x_i \in \{0, 1\}$, $i = 1, \dots, n$ let

$$\Pr\{X_1 = x_1, \dots, X_n = x_n\} = p(x_1, \dots, x_n)$$

give the Probability Mass Function (PMF) for $X_1, \dots, X_n$.

- (i) What does it mean to say that the sequence is an infinite exchangeable sequence?

Solution: (2 marks, bookwork) The sequence of random variables $X_1, X_2, X_3, \dots$ is exchangeable if and only if, for every $n \geq 1$,

$$p(x_1, x_2, \dots, x_n) = p(x_{\sigma_1}, x_{\sigma_2}, \dots, x_{\sigma_n})$$

for every permutation $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n)$ of the numbers $(1, 2, \dots, n)$.

- (ii) Show that there exists a generative model for $X_1, X_2, X_3, \dots$ given in terms of a cumulative distribution function $F : [0, 1] \rightarrow [0, 1]$, in which

$$Q \sim F \text{ and, independently for } i = 1, 2, 3, \dots, X_i \sim \text{Bernoulli}(Q).$$

You may quote de Finetti's theorem without proof.

Solution: (2 marks, bookwork) de Finetti for an infinite exchangeable sequence of binary random variables: There exists a CDF $F(q)$ on $[0, 1]$ such that

$$p(x_1, \dots, x_n) = \int_0^1 q^{x_i} (1 - q)^{1-x_i} dF(q).$$

The random variable

$$Q = \lim_{N \rightarrow \infty} N^{-1} \sum_{i=1}^N X_i \text{ exists and } F(q) = \Pr(Q \leq q).$$

Now if we take $Q \sim F$ and (when $Q = q$) $X_i \sim \text{Bernoulli}(q)$, $i = 1, 2, 3, \dots$ jointly independent, then

$$\Pr(X_1 = x_1, \dots, X_n = x_n) = \int_0^1 \left[\prod_{i=1}^n p(x_i|q) \right] dF(q)$$

so this generative model reproduces the PMF given by de Finetti, so it generates the X 's.

- (iii) Suppose that, for some fixed $0 < a < 1$ and $k(x) = \sum_{i=1}^n x_i$, we set

$$p(x_1, \dots, x_n) = \frac{a^{k(x)}(1-a)^{n-k(x)}}{3} + \frac{4}{3} \frac{\Gamma(k(x)+2)\Gamma(n-k(x)+1)}{\Gamma(n+3)}.$$

Give an explicit generative model for the infinite exchangeable sequence $X_1, X_2, X_3, \dots$ admitting $p(x_1, \dots, x_n)$ as its marginal PMF for every $n \geq 1$.

Solution: (5 marks, new) Looking at the de Finetti form

$$p(x_1, \dots, x_n) = \int_0^1 q^{k(x)} (1 - q)^{n-k(x)} dF(q)$$

we see we get the first term if we take a density $dF(q) = f(q)dq$ of the form

$$f(q) = \delta_a(q)/3 + 2g(q)/3$$

and $g(q)$ a density giving the second term,

$$\int_0^1 q^k (1-q)^{n-k} g(q) dq = 2 \times \frac{\Gamma(k+2)\Gamma(n-k+1)}{\Gamma(n+3)}.$$

Now

$$\int_0^1 q^k (1-q)^{n-k} dq = \Gamma(k+1)\Gamma(n-k+1)/\Gamma(n+2),$$

so we get what we want if $g(q) = 2q$, a normalised density in $[0, 1]$. The density we want is then

$$f(q) = \delta_a(q)/3 + 4q/3$$

so the CDF is

$$F(q) = \begin{cases} 0 & \text{for } q < 0 \\ \frac{1}{3}\mathbb{I}_{q \geq a} + \frac{2}{3}q^2 & \text{for } 0 \leq q \leq 1 \\ 1 & \text{for } q > 1 \end{cases}.$$

The generative model is as stated in the question with this choice of F .

- (b) [11 marks] Let $G = (V, E)$ be a directed graph with nodes $V = \{1, \dots, n\}$ and edges $E \subseteq V \times V$. A Directed Acyclic Graph (DAG) is a directed graph with no directed loops. A walk following the directed edges cannot return to its starting node.

For $n = 1, 2, 3, \dots$ let $\mathcal{G}_n$ be the set of all distinct DAG's on n labeled nodes and let $|\mathcal{G}_n|$ give the number of such DAG's. There are three DAG's on $n = 2$ nodes with $V = \{1, 2\}$ and $G_i = (V, E_i)$, $i = 1, 2, 3$. Possible edge sets are $E_1 = \emptyset$, $E_2 = \{(1, 2)\}$ and $E_3 = \{(2, 1)\}$, so $|\mathcal{G}_2| = 3$. There are $|\mathcal{G}_3| = 25$ DAG's on $n = 3$ nodes.

- (i) Let $\pi_{U,n}(g)$, $g \in \mathcal{G}_n$ give the uniform distribution on DAG's with n nodes, so that

$$\pi_{U,n}(g) = \frac{\mathbb{I}_{g \in \mathcal{G}_n}}{|\mathcal{G}_n|}.$$

For $G \sim \pi_{U,3}$ and $G = (V, E)$ let $G_{-3} \in \mathcal{G}_2$ be the DAG obtained from G by removing node 3 and any edge $(i, 3) \in E$ or $(3, i) \in E$, $i = 1, 2$ connected to node 3. Show that G_{-3} is not distributed according to $\pi_{U,2}$.

Solution: (3 marks, standard) 25 into 3 wont go. Let $g \in \mathcal{G}_2$ be the DAG $g = (V = \{1, 2\}, E = \emptyset)$ so $\pi_{U,2}(g) = 1/3$. It is sufficient to show that

$$\pi_{U,2}(g) \neq \sum_{g' \in \mathcal{G}_3} \pi_{U,n}(g') \mathbb{I}_{g'_{-3}=g}.$$

The LHS is equal $1/3$. The set $\{g' \in \mathcal{G}_3 : g'_{-3} = g\}$ has 9 elements (one with 0 edges, 4 with one edge and 4 with two edges) so since $1/3 \neq 9/25$ we have $\Pr(G_{-3} = g) \neq \pi_{U,2}(g)$. We dont actually have to count DAGs in $\{g' \in \mathcal{G}_3 : g'_{-3} = g\}$ as there is no integer m such that $m/25 = 1/3$.

- (ii) Let $\mathcal{S}$ be the set of all subsets of $\mathcal{G}_2 \cup \mathcal{G}_3$. For $G \in \mathcal{G}_3$ and $A \in \mathcal{S}$ event A occurs if $G \in A$ or $G_{-3} \in A$. Explain briefly why there does not exist a probability

distribution over $\mathcal{G}_3$ expressing prior preferences over sets $A \in \mathcal{S}$ which match the prior preferences expressed by $\pi_{U,n}$, $n = 2, 3$.

Solution: (3 marks, new) If such a distribution, π say, existed it would satisfy the axioms of probability. The axioms of probability imply marginal consistency, so if $G \sim \pi$ and we take $A = \{g\}$ with $g = (V = \{1, 2\}, E = \emptyset)$ and $B = \{g' \in \mathcal{G}_3 : g'_{-3} = g\}$ we must have $A \sim B$. But we have seen (in the answer to the last question) that $A < B$ in the preferences expressed by $\pi_{U,n}$, $n = 2, 3$ (again it is enough that $A \not\sim B$).

(iii) Suppose for g a graph on n nodes we have a function $\text{is.DAG}(g)$,

$$\text{is.DAG}(g) = \begin{cases} 1 & \text{if } g \in \mathcal{G}_n \\ 0 & \text{otherwise,} \end{cases}$$

so $\text{is.DAG}(g)$ checks g is a DAG. Give an MCMC algorithm targeting $\pi_{U,n}$ for fixed $n \geq 2$. Explain briefly why your MCMC chain is ergodic.

Solution: (5 marks, new) Let $X_t = g$ with $g \in \mathcal{G}_n$ and $g = (V, E)$.

Step 1. Pick an ordered pair of vertices $(i, j) \in V \times V$ uniformly at random. If $(i, j) \in E$ then set $E' = E \setminus \{(i, j)\}$ and if $(i, j) \notin E$ then set $E' = E \cup \{(i, j)\}$. Set $g' = (V, E')$.

Step 2 If $\text{is.DAG}(g') = 1$ then set $X_{t+1} = g'$ and otherwise set $X_{t+1} = g$.

This works because if $g, g' \in \mathcal{G}_n$ and $q(g'|g)$ is the probability to propose g' given the current state is g then

$$q(g'|g) = \begin{cases} 1/n^2 & \text{if } |E' \Delta E| = 1 \\ 0 & \text{otherwise,} \end{cases}$$

where $E' \Delta E = (E' \setminus E) \cup (E \setminus E')$ and $|A|$ is the number of elements in the set A . The acceptance probability is

$$\begin{aligned} \alpha(g'|g) &= \min \left\{ 1, \frac{\pi_{U,n}(g')q(g|g')}{\pi_{U,n}(g)q(g'|g)} \text{is.DAG}(g') \right\} \\ &= \text{is.DAG}(g'). \end{aligned}$$

The chain is irreducible. If we pick a pair of states g, g' then we can delete from g down to the empty DAG $(V, \emptyset)$ and then add the edges in g' one at a time without creating a cycle at any step. It is aperiodic as it is irreducible and we can reject. The space of states $\mathcal{G}_n$ is finite, so it is ergodic.